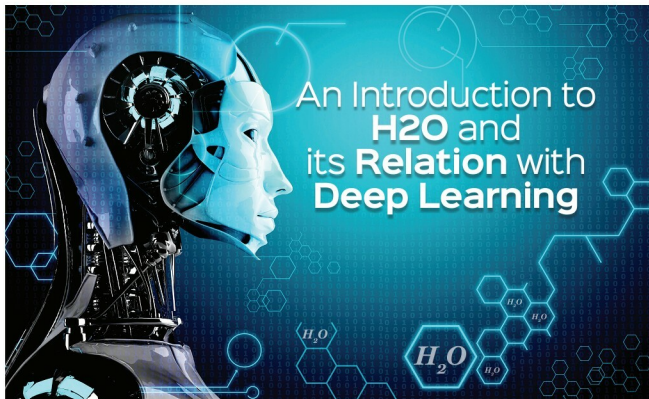




H2O, produced by the start up H2O.ai (formerly Oxdata), was launched in Silicon Valley in 2011. The speed of this open source software for big data analytics allows users to fit thousands of models for pattern discovery in data. H2O is written in Java, Python and R, and has many useful features on offer for deep learning.

Deep learning, defined simply, is ‘a class of machine learning techniques, where many layers of information processing stages in hierarchical supervised architectures are exploited for unsupervised feature learning and for pattern analysis/classification’. The essence of deep learning is to compute hierarchical features or representations of observational data, where higher-level features or factors are defined from lower-level ones. Although there are many similar definitions and architectures for deep learning, two common elements in all of them are: multiple layers of non-linear information-processing, and supervised or unsupervised learning of feature representations at each layer, from the features learned at the previous layer. The initial work on deep learning was based on multi-layer neural network models. Recently, many other models are also being used, such as deep kernel machines and deep Q-networks.

Introducing H2O

According to Oxdata (its creator), H2O is ‘the open source in-memory, prediction engine for Big Data science’. H2O is a feature-rich, open source machine learning platform known for its R and Spark integration and its ease of use. It is a Java virtual machine that is optimised for doing in-memory processing of distributed, parallel machine learning

algorithms on clusters. A cluster is a software construct that can be fired up on your laptop, on a server, or across the multiple nodes of a cluster of real machines, including computers that form a Hadoop cluster.

Deep learning for IoT using H2O

In a nutshell, deep learning methods implement neural network algorithms such as the Feed Forward Back Propagation Neural Network, Convolved Neural Networks, Recursive Neural Networks and others.

Deep learning algorithms play an important role in IoT analytics. Data from machines is sparse and/or has a temporal element in it. Even when we trust data from a specific device, it may behave differently under different conditions. Hence, capturing all scenarios for the data pre-processing/training stage of an algorithm is difficult. Monitoring sensor data continuously is also cumbersome and expensive. Deep learning algorithms can help to mitigate these issues. These algorithms learn on their own, allowing the developer to concentrate on more important tasks without worrying about training them.

Let’s now discuss what H2O offers as a part of its deep learning framework, and the features that make it suitable for data from things.